

BPHO ROUND 0 · MARKSCHEME

Drill bank section 6

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking

Original questions written for this course, in the Round 0 format. Not BPhO material.

Quick key

Q1 A	Q2 C	Q3 B	Q4 D	Q5 E
Q6 B	Q7 A	Q8 C	Q9 E	Q10 D
Q11 B	Q12 A	Q13 E	Q14 C	Q15 D
Q16 A	Q17 E	Q18 B	Q19 C	Q20 D
Q21 A	Q22 C	Q23 B	Q24 E	Q25 D

Worked solutions

Q1 · A voltmeter of finite resistance loads the potential divider it is measuring, so the reading is below the unloaded value — **Answer A**

What is being tested. Whether you treat the voltmeter as part of the circuit. A voltmeter of finite resistance takes current, and that current has to come through the $4.0\ \Omega$ resistor, which means less voltage is left for the branch being measured.

Step 1 — the meter changes the branch it is across. Before the meter is connected, the branch is $8.0\ \Omega$. With the meter across it, the branch is the $8.0\ \Omega$ and the $24\ \Omega$ in parallel. That combination is smaller than either, which is the whole point.

Step 2 — combine the pair.

$$R_{\text{pair}} = 8.0 \times 24 / (8.0 + 24) = 192/32 = 6.0\ \Omega$$

Step 3 — the circuit resistance. The pair is in series with the $4.0\ \Omega$:

$$R_{\text{total}} = 4.0 + 6.0 = 10\ \Omega$$

Step 4 — the supply current.

$$I = 12/10 = 1.2\ \text{A}$$

Compare that with the unloaded current, $12/12 = 1.0\ \text{A}$. The meter has made the circuit draw *more* current, not less: it added a path, so the total resistance fell from $12\ \Omega$ to $10\ \Omega$.

Step 5 — the reading. The meter reads the p.d. across the pair it is connected to:

$$V = I \times R_{\text{pair}} = 1.2 \times 6.0 = 7.2\ \text{V}$$

Sanity check. The reading is 7.2 V, which is below the unloaded 8.0 V. Loading must always pull the reading towards zero, never above it, so the direction of the change is right. A useful cross-check: the current through the $8.0\ \Omega$ is $7.2/8.0 = 0.90\text{ A}$ and the current through the meter is $7.2/24 = 0.30\text{ A}$, and $0.90 + 0.30 = 1.2\text{ A}$, which is the supply current from Step 4.

The other four options are the readings four different mistakes give. Reporting 8.0 V means never connecting the meter at all. Reporting 6.0 V means reading the pair's resistance in ohms as if it were a voltage. Reporting 4.8 V means reading across the wrong resistor. Reporting 2.4 V means multiplying the meter's current by the $8.0\ \Omega$ instead of by the meter's own resistance.

Answer: A

The trap. The classic error is to answer with the unloaded divider value, 8.0 V, and never notice that the meter is a resistor. The second trap is subtler: the supply current RISES when the meter is connected (1.0 A becomes 1.2 A), so a candidate who assumes the meter can only reduce the current reaches a wrong number by a different route.

Q2 · Propagating a small fractional change through a square root, using the first-order expansion — **Answer C**

What is being tested. Whether you propagate a small change through a power law, rather than substituting numbers and doing the arithmetic. The 4.0 % is a hint: it is small, so the first-order expansion is the intended tool.

Step 1 — write the dependence. The period of a simple pendulum is

$$T = 2\pi \sqrt{\ell/g}$$

Everything in front of the square root is constant, so the question is how the square root of ℓ responds to a 4.0 % change in ℓ .

Step 2 — write the change as a factor. If the length becomes $\ell(1 + x)$ with $x = 0.040$, the new period is

$$T_{\text{new}} = T \times (1 + x) \text{ to the power one half}$$

Step 3 — expand to first order. For a small x ,

$$(1 + x) \text{ to the power one half} \approx 1 + x/2$$

This is the small-parameter approximation the BPhO constant sheet supplies, and the reason it supplies it is that questions like this are set. It is valid here because 0.040 is far smaller than 1.

Step 4 — read off the change. The new period is $T(1 + 0.020)$, so the fractional change is

$$\Delta T/T \approx x/2 = 0.040/2 = 0.020$$

which is 2.0 %.

Sanity check. A longer pendulum has a longer period, so the change must be an increase, and it is. And the change must be smaller than the 4.0 % that caused it, because a square root always shrinks a small fractional change; a quarter power would shrink it further, and a square would enlarge it.

Notice that the answer does not depend on g , on the original length, or on the original period. That is a good sign: a question that asks only for a percentage change should not need the absolute values, and if

your method needs them you have made it harder than it is.

Answer: C

The trap. Two traps. The first is treating the period as proportional to the length, which gives 4.0 % and is the answer for a stretched spring rather than a pendulum. The second is remembering that a square root halves the change but applying the half in the wrong direction, which gives 8.0 %.

Q3 · A block pulled up a rough inclined plane at constant speed: resolving along the plane and getting the direction of friction right — **Answer B**

What is being tested. Two things at once: whether you resolve the weight correctly, and whether you get the direction of friction right. The figure settles the second, so a wrong answer here is a wrong reading of the figure.

Step 1 — resolve the weight. The weight is

$$W = mg = 2.0 \times 10 = 20 \text{ N}$$

On a plane inclined at 30° , the component down the slope is $W \sin 30$ and the component perpendicular to the slope is $W \cos 30$. This is the step most often reversed: it is the *along* component that carries the sine, because at zero inclination the along component must vanish.

$$W_{\text{along}} = 20 \sin 30 = 10 \text{ N} \quad W_{\text{perp}} = 20 \cos 30 = 17.3 \text{ N}$$

The perpendicular component needs $\cos 30$, and $\cos 30 = \sqrt{3}/2$. With $\sqrt{3} \approx 1.73$ that is 0.87, so $20 \cos 30 \approx 17.3 \text{ N}$. $\sqrt{3}$ to two decimal places is one of the few numerical facts a non-calculator paper expects you to carry.

Step 2 — the normal reaction. There is no acceleration perpendicular to the plane, so the normal reaction balances the perpendicular component. It is *not* the whole weight:

$$N = 20 \cos 30 = 17.3 \text{ N}$$

This is the step that separates 14 N from 15 N. Using $N = W = 20 \text{ N}$ would give a friction of 5.0 N.

Step 3 — the friction.

$$f = \mu N = 0.25 \times 17.3 = 4.3 \text{ N}$$

Step 4 — which way does the friction point? The block is moving *up* the slope, so friction acts *down* it. That is what the green arrow in the figure shows. So along the plane there are two forces opposing the motion and one driving it.

Step 5 — balance along the plane. The speed is constant, so there is no acceleration and the forces along the plane must balance:

$$F = W_{\text{along}} + f = 10 + 4.3 = 14.3 \text{ N}$$

To two significant figures, F is 14 N.

A sanity check on the alternatives. If the friction were drawn up the slope — the error you make by thinking of the block as sliding down — you would subtract and get $10 - 4.3 = 5.7 \text{ N}$, which is less than

the weight component alone and therefore cannot move the block up the slope at all. That is a physical argument that rules the option out without doing any arithmetic, and it is worth noticing that the paper rewards noticing it.

Answer: B

The trap. The trap is the direction of friction. A block sliding DOWN a slope feels friction UP it, and candidates carry that picture over. Here the block is pulled up, so friction points down and the two forces add. A second trap is using the whole weight as the normal reaction, which gives 15 N.

Q4 · A circuit with a nested parallel pair and internal resistance: working from the supply down to the current in one innermost resistor — **Answer D**

What is being tested. A reduction long enough that it cannot be done in one step, and a question that asks for something deep inside it. The numbers are chosen so that every stage is exact, which is what makes it a non-calculator question.

Step 1 — the innermost pair. The $6.0\ \Omega$ and the $3.0\ \Omega$ are in parallel:

$$6.0 \times 3.0 / (6.0 + 3.0) = 18/9 = 2.0\ \Omega$$

Step 2 — the second branch. That pair is in series with the $4.0\ \Omega$:

$$4.0 + 2.0 = 6.0\ \Omega$$

Step 3 — the parallel section. The second branch is $6.0\ \Omega$ and the first branch is also $6.0\ \Omega$, and they are in parallel:

$$6.0 \times 6.0 / (6.0 + 6.0) = 3.0\ \Omega$$

Step 4 — the circuit resistance. The parallel section is in series with the $5.0\ \Omega$ and with the cell's own $2.0\ \Omega$:

$$R = 2.0 + 5.0 + 3.0 = 10\ \Omega$$

The internal resistance must be included. It carries the whole supply current, so leaving it out would overstate every current in the circuit by twenty per cent.

Step 5 — the supply current.

$$I = \mathcal{E}/R = 12/10 = 1.2\ \text{A}$$

Step 6 — the p.d. across the parallel section.

$$V_{\text{section}} = 1.2 \times 3.0 = 3.6\ \text{V}$$

Step 7 — the current in the second branch. Both branches sit across $3.6\ \text{V}$, and the second branch is $6.0\ \Omega$:

$$I_{\text{branch}} = 3.6/6.0 = 0.60\ \text{A}$$

As a check, the first branch is $3.6/6.0 = 0.60$ A as well, and $0.60 + 0.60 = 1.2$ A, which is the supply current. The two branches are equal, so the supply current splits equally, and that is a good sign the reduction was right.

Step 8 — the p.d. across the innermost pair. Within the second branch, the $4.0\ \Omega$ takes $0.60 \times 4.0 = 2.4$ V, so the pair takes the rest:

$$V_{\text{pair}} = 0.60 \times 2.0 = 1.2\ \text{V}$$

Step 9 — the current in the $3.0\ \Omega$ resistor. The pair sits across 1.2 V, and the $3.0\ \Omega$ is one arm of it:

$$I = 1.2/3.0 = 0.40\ \text{A}$$

The other arm carries $1.2/6.0 = 0.20$ A, and $0.40 + 0.20 = 0.60$ A, which is the branch current. Every junction adds up, which is the check that the whole chain of p.d.s was walked correctly.

Step 10 — the power.

$$P = I^2 R = 0.40 \times 0.40 \times 3.0 = 0.48\ \text{W}$$

Notice how easy it is to answer for the wrong resistor: the $6.0\ \Omega$ in the pair dissipates $0.20^2 \times 6.0 = 0.24$ W, and the branch current 0.60 A applied to the $4.0\ \Omega$ gives 1.4 W. The question names the $3.0\ \Omega$, and the only way to be sure you have it is to walk the chain as above.

Answer: D

The trap. Two traps. Leaving out the internal resistance gives $12/(5 + 3) = 1.5$ A instead of 1.2 A, and every later number is then wrong. And answering for the neighbouring resistor is easy: the $6.0\ \Omega$ in the same pair gives 0.24 W, which is one of the options. The question asks for the $3.0\ \Omega$, so the last division matters.

Q5 · An order-of-magnitude estimate assembled from a rate, a unit conversion and a lifetime — **Answer E**

What is being tested. Whether you can assemble an estimate from two quantities you have to supply yourself, and whether you know that "to the nearest power of ten" is a rounding decision rather than a guess.

Step 1 — what is the answer the product of? Beats in a lifetime is a rate multiplied by a duration. Nothing else enters. Writing that down first is what stops the estimate drifting.

$$\text{beats} = \text{rate} \times \text{lifetime}$$

Step 2 — the rate, in beats per year. The rate is given per minute, so the conversion needs the number of minutes in a year:

$$60 \times 24 \times 365 \approx 60 \times 24 \times 400 = 5.8 \times 10 \text{ to the 5 minutes a year}$$

Rounding the year to 400 days is a five per cent error, which is far below the precision the answer needs. Then

$$70 \times 5.8 \times 10 \text{ to the 5} \approx 4 \times 10 \text{ to the 7 beats a year}$$

Step 3 — the lifetime. Eighty years is the usual round figure, and the exact value does not matter: fifty years and a hundred years differ by a factor of two, which is nowhere near a factor of ten.

Step 4 — multiply.

$$4 \times 10 \text{ to the } 7 \times 80 = 3.2 \times 10 \text{ to the } 9$$

Step 5 — round to the nearest power of ten. The boundary between two neighbouring powers of ten is not halfway in the ordinary sense. They are separated by a factor of ten, so the dividing line sits at $\sqrt{10} \times 10 \text{ to the } 9$. Since $\sqrt{10} \approx 3.16$, the line is at $3.16 \times 10 \text{ to the } 9$, and the question is whether the estimate falls above it or below. Carrying the arithmetic exactly:

$$70 \times 60 \times 24 \times 365 \times 80 = 2.9 \times 10 \text{ to the } 9$$

2.9 is below 3.16, so the answer is 10 to the 9. It is worth seeing how close to the line this sits: the rounded route above gave $3.2 \times 10 \text{ to the } 9$, which is on the far side of 3.16 and so tips over to 10 to the 10. That is the honest lesson of the question. An estimate of this kind is good to within a factor of a few, and near a boundary the factor of a few decides the answer.

Answer: E

The trap. The trap is the rounding rule. $3 \times 10 \text{ to the } 9$ looks closer to 10 to the 10 than to 10 to the 9 if you compare the leading digit, but 3 is closer to 1 than to 10 in the multiplicative sense that a power-of-ten scale uses. The logarithm decides, not the digit.

Q6 · Average velocity from a velocity-time graph: the area under the graph divided by the whole time, expressed as a fraction of the peak speed — **Answer B**

What is being tested. Whether you use the area under the graph rather than the peak, and whether you can carry the peak through as a symbol so that the arithmetic stays clean.

Step 1 — the rule. Average velocity is the total displacement over the total time. The displacement is the area under a velocity-time graph, and the total time is the whole 12 s, including the parts where the cyclist is slow.

$$v_{\text{avg}} = \text{area} / \text{total time}$$

Step 2 — split the area. The shape is a trapezium with a flat top, which splits into two triangles and a rectangle. The cyclist's speed is approximately constant between 4.0 s and 10 s, so the middle region is a rectangle rather than a curve.

Step 3 — each area in terms of v.

$$\begin{aligned} \text{triangle 1: } & 0.5 \times 4.0 \times v = 2v \\ \text{rectangle: } & 6.0 \times v = 6v \\ \text{triangle 2: } & 0.5 \times 2.0 \times v = v \end{aligned}$$

Each triangle contributes half of the rectangle that would enclose it. That is the discriminator of the figure: a candidate who counts each ramp as a rectangle gets $4v + 6v + 2v = 12v$, and $12v/12 \text{ s} = v$, which is the last option.

Step 4 — add and divide.

$$\begin{aligned}\text{total area} &= 2v + 6v + v = 9v \\ v_{\text{avg}} &= 9v / 12 = 3v/4\end{aligned}$$

Sanity check. Three quarters of the peak is below the peak, which it must be, because the cyclist is below the peak for four of the twelve seconds. And it is well above a half, because the cyclist spends half the interval at the full peak.

With the numbers put in, the displacement is $9 \times 8.0 = 72 \text{ m}$ and the average velocity is $72/12 = 6.0 \text{ m s}^{-1}$. Notice that the question did not need the 8.0: the answer is a fraction of v , and the fraction is the physics.

Answer: B

The trap. The trap is treating the average as the peak, which happens whenever the two ramps are counted as rectangles. The second trap is dividing by the time spent at the peak rather than by the whole interval: 72 m over 10 s gives 7.2 m s^{-1} , which is also an option in a variant of this question.

Q7 · An elastic collision between equal masses, one at rest, where the two outgoing directions are symmetric: finding the angle — **Answer A**

What is being tested. Whether you use both conservation laws and then eliminate the unknown speed between them. Using only one of them leaves an equation with two unknowns, which is where most candidates stop.

Step 1 — resolve the momentum. Momentum is a vector, so conserving it means conserving each component separately. Take the original direction as the first axis and the perpendicular as the second.

Step 2 — the perpendicular equation. The incoming particle has no perpendicular momentum. The two outgoing particles have equal masses and equal speeds, and they emerge on opposite sides, so their perpendicular momenta are equal and opposite:

$$mv \sin \theta - mv \sin \theta = 0$$

This is satisfied for any θ . It tells us nothing about the angle — but it does confirm that the symmetry assumed in the question is self-consistent, which is worth knowing before relying on it.

Step 3 — the momentum along the original direction. Each outgoing particle contributes $v \cos \theta$ along the original direction:

$$mu = 2mv \cos \theta \quad \text{so} \quad u = 2v \cos \theta$$

Step 4 — the kinetic energy. The collision is elastic, so

$$\text{one half } m u^2 = 2 \times \text{one half } m v^2 \quad \text{so} \quad u^2 = 2v^2 \quad \text{and} \quad u = \sqrt{2} v$$

Step 5 — divide to eliminate v . Substitute $u = \sqrt{2} v$ into the momentum equation:

$$\sqrt{2} v = 2v \cos \theta \quad \text{so} \quad \cos \theta = \sqrt{2}/2$$

The speed v has cancelled, which is the whole point of using both laws: neither equation alone fixes the angle, but together they do, because the angle appears in one and not the other.

Step 6 — the angle. The value $\sqrt{2}/2$ is the cosine of 45° , and it is worth knowing by heart, because 45° is the one angle for which the sine and the cosine are equal. So $\theta = 45^\circ$.

A check. Put the angle back into both laws. With $\theta = 45^\circ$ the momentum equation gives $u = 2v \cos 45^\circ = 2v \times (\sqrt{2}/2) = \sqrt{2} v$, so $v = u/\sqrt{2}$. The kinetic energy afterwards is $2 \times \text{one half } m (u/\sqrt{2})^2 = m u^2/2$, which is the initial kinetic energy. The momentum along the original direction is $2m \times (u/\sqrt{2}) \times (1/\sqrt{2}) = mu$, which is the initial momentum. Both laws are satisfied, so the answer is consistent.

This is the general result for equal masses with one at rest: the two outgoing paths are perpendicular. The symmetry imposed in this question then puts each of them at 45° to the original direction, which is the special case of that general rule.

Answer: A

The trap. The trap is to remember that the outgoing paths are perpendicular and then to report 90 degrees as the angle. The perpendicularity is a statement about the SUM of the two angles, not about either one of them, and here the symmetry splits the 90 degrees evenly.

Q8 · Young's modulus: finding a load from a measured extension, using the small-strain assumption — Answer C

What is being tested. Whether you write the strain with the original length, and whether you notice how small it is. The smallness is not decoration: it is what makes the answer a single multiplication.

Step 1 — write the definition in full. Young's modulus is

$$E = \text{stress} / \text{strain} = (F/A) / (\Delta\ell/\ell)$$

The strain is the extension divided by the *original* length. Dividing by the stretched length would be wrong, though here the two differ by so little that it would not change the answer — which is worth knowing, because it means you should not spend time worrying about it.

Step 2 — the strain. Convert the extension to metres first:

$$\begin{aligned}\Delta\ell &= 1.0 \text{ mm} = 1.0 \times 10^{-3} \text{ m} \\ \text{strain} &= 1.0 \times 10^{-3} / 2.0 = 5.0 \times 10^{-4}\end{aligned}$$

A strain of five parts in ten thousand. The wire has stretched by 0.05 % of its length. This is the small-strain regime that the whole theory of Young's modulus assumes, and it is why the cross-sectional area may be taken as its original value and the wire's own weight negligible beside the load.

Step 3 — the stress. Rearranging the definition:

$$\text{stress} = E \times \text{strain} = 2.0 \times 10^{11} \times 5.0 \times 10^{-4} = 1.0 \times 10^8 \text{ Pa}$$

That is 100 MPa, which is a sensible working stress for steel — well below its breaking stress of several hundred MPa, so the wire is in no danger of yielding.

Step 4 — the force, then the mass.

$$\begin{aligned}F &= \text{stress} \times A = 1.0 \times 10^8 \times 1.0 \times 10^{-6} = 100 \text{ N} \\ m &= F/g = 100/10 = 10 \text{ kg}\end{aligned}$$

Sanity check. A 10 kg mass is a bucket of water, and it stretches a one square millimetre steel wire by a millimetre over two metres. That feels right: steel is stiff, and a millimetre is a very small extension for a wire two metres long. If your answer had come out as a tonne, the strain would have had to be a hundred times larger, which the given extension rules out.

Answer: C

The trap. The trap is the unit of the extension. The strain is a ratio, and a ratio of lengths is dimensionless, so a candidate who leaves the extension in millimetres and the length in metres gets a strain a thousand times too small and a mass of 0.010 kg. Converting to metres before dividing is the whole of the discipline.

Q9 · Superposition of two pulses of opposite polarity, expressed as a fraction of the larger amplitude — **Answer E**

What is being tested. Whether you carry the sign of a displacement through the addition. The amplitudes are 3.0 cm and 1.0 cm, and the only thing that decides the answer is that one of them is negative.

Step 1 — when do they meet? Both pulses travel towards the midpoint, and the figure marks it with a dashed line. Since the pulses are drawn equidistant from it, they arrive together. The question tells you to take the instant of full overlap, so no timing arithmetic is needed.

Step 2 — assign signs. Take upward as positive. Then the left-hand pulse contributes

$$y_1 = +3.0 \text{ cm}$$

and the right-hand pulse contributes

$$y_2 = -1.0 \text{ cm}$$

Writing the second one down as negative is the whole of the physics. A candidate who writes both as positive is adding two amplitudes that point in opposite directions.

Step 3 — add them.

$$y = y_1 + y_2 = 3.0 - 1.0 = +2.0 \text{ cm}$$

The string is 2.0 cm above its rest position at the midpoint. It is not flat, and it is not 4.0 cm up.

Step 4 — as a fraction of A. The larger amplitude is $A = 3.0 \text{ cm}$, so

$$y = 2.0/3.0 \times A = 2A/3$$

Step 5 — what happens next. The pulses do not bounce off one another. After the instant of overlap each continues in its original direction with its original shape, and a moment later the midpoint is flat again while two separate pulses travel on. Superposition is instantaneous and leaves no trace.

The three wrong answers are the three ways of getting the sign wrong. Reporting zero assumes the pulses are equal and opposite. Reporting A ignores the smaller pulse. Reporting $4A/3$ adds the magnitudes, which would be right only if both pulses displaced the string upward.

Answer: E

The trap. The trap is the sign. The amplitudes 3.0 cm and 1.0 cm look like a simple subtraction problem, and the option list contains the sum $4A/3$ as a temptation for anyone who adds the numbers and then has to decide about the sign afterwards.

Q10 · Refraction at a plane surface: finding the speed of light in the material from the two angles, with both measured from the normal — **Answer D**

What is being tested. Whether you use the sines rather than the angles, and whether you remember which way the refractive index goes. Both angles in the figure are measured from the normal, which is the convention Snell's law needs.

Step 1 — Snell's law. With air taken as having a refractive index of one:

$$n = \sin i / \sin r$$

Step 2 — the sines. Both angles are known exactly:

$$\sin 60^\circ = \sqrt{3}/2 = 0.866 \quad \sin 30^\circ = 1/2 = 0.500$$

If you measure the angles from the surface instead of from the normal, the roles are exchanged and you would compute $0.500/0.866 = 0.58$, which is a refractive index below one and therefore impossible for a material.

Step 3 — the refractive index.

$$n = (\sqrt{3}/2) / (1/2) = \sqrt{3} = 1.7$$

Step 4 — the definition in terms of speed. The refractive index is the ratio of the speed of light in vacuum to the speed in the material:

$$n = c/v$$

Step 5 — the speed. Rearranging:

$$v = c/n = 3.0 \times 10^8 / 1.7 = 1.7 \times 10^8 \text{ m s}^{-1}$$

Sanity check. Light entering a material from air slows down, so the answer must be below $3.0 \times 10^8 \text{ m s}^{-1}$. It is, and by a factor of $\sqrt{3}$. A speed above that of light in vacuum would be a contradiction, which rules out the 5.2 option without any arithmetic.

Notice the coincidence of the numbers: the refractive index is 1.7 and the speed is $1.7 \times 10^8 \text{ m s}^{-1}$. That happens because the speed of light is 3.0×10^8 and $\sqrt{3}$ is close to 1.7. It is worth being able to see that it is a coincidence rather than a rule.

Answer: D

The trap. Two traps. Measuring the angles from the surface rather than the normal inverts the ratio and gives a refractive index of 0.58, which is physically impossible. And multiplying by the refractive index rather than dividing gives a speed greater than that of light in vacuum, which is the 5.2 option.

Q11 · Power dissipated inside a cell, obtained symbolically by eliminating the current between Ohm's law and the power relation — **Answer B**

What is being tested. Whether you can produce an expression rather than a number, and whether you square the current. Every option here is a plausible-looking arrangement of the same four symbols, so recognising one is not enough: it has to be derived.

Step 1 — the current. The internal resistance is in series with the external resistor, so both carry the same current and the total resistance is $R + r$. Ohm's law for the whole circuit gives

$$I = \varepsilon / (R + r)$$

This is the only current in the circuit. There is no separate current through the cell.

Step 2 — the power in a resistor.

$$P = I^2 R_{\text{component}}$$

The square is not optional. It comes from combining $P = VI$ with $V = IR$, and dropping it is the single most common error in this question.

Step 3 — substitute and collect. The component is the internal resistance r :

$$P_{\text{internal}} = (\varepsilon / (R + r))^2 \times r = \varepsilon^2 r / (R + r)^2$$

Step 4 — check the dimensions. ε^2 has the dimensions of volts squared and $(R + r)^2$ has ohms squared, so the ratio is volts squared over ohms. Since $V = IR$, volts squared over ohms is volts times amperes, which is watts. The expression is a power. The option $\varepsilon^2 r / (R + r)$ has an extra ohm in the numerator, which makes it a power times a resistance, so it cannot be right.

Sanity check. If r were zero there would be no energy dissipated inside the cell, and the expression correctly gives zero. If R were zero the cell would be short-circuited, and the expression gives ε^2 / r , which is the largest power the cell can dissipate internally. Both limits are sensible, which is what you want from an expression you cannot evaluate numerically.

For comparison, the power in the external resistor is $\varepsilon^2 R / (R + r)^2$, which is the same shape with R in the numerator. That is the fourth option, and it answers a different question.

Answer: B

The trap. The trap is the missing square. The option without it, $\varepsilon^2 r / (R + r)$, has the dimensions of a power multiplied by a resistance, which is not a power at all, so a dimensional check settles it. The other trap is answering for the external resistor, which is the same expression with R in the numerator.

Q12 · The number of nuclei decaying in an interval short compared with the half-life, using the constant-rate approximation — **Answer A**

What is being tested. The difference between an activity and a count. The activity is a rate per second; the number decaying is that rate multiplied by the time, and only if the rate is effectively constant over the interval.

Step 1 — the activity. The activity is the decay constant times the number of nuclei present:

$$A = \lambda N = 1.0 \times 10^{-8} \times 4.0 \times 10^{20} = 4.0 \times 10^{12} \text{ s}^{-1}$$

So 4.0×10^{12} nuclei decay every second. That is the rate, not the answer.

Step 2 — is the rate constant? The half-life is

$$t_{1/2} = \ln 2 / \lambda = 0.69 / 1.0 \times 10^{-8} \approx 6.9 \times 10^7 \text{ s}$$

which is about two years. The interval we are asked about is 60 s, which is smaller than the half-life by a factor of about a million. So the population is unchanged to six significant figures over the interval, and the activity may be treated as constant.

This check is the reason the word "approximately" appears in the question. For an interval comparable with the half-life the constant-rate shortcut would be wrong, and the exponential would have to be used.

Step 3 — the count.

$$\text{number decaying} = A \times t = 4.0 \times 10^{12} \times 60 = 2.4 \times 10^{14}$$

Sanity check. The number decaying, 2.4×10^{14} , is 6.0×10^{-7} of the 4.0×10^{20} present. That is a millionth, which is exactly what a 60 s interval out of a two-year half-life should give. If the count had come out comparable with the population, the approximation would have been self-contradictory and the answer would have to be reworked.

Answer: A

The trap. The trap is stopping at the activity. 4.0×10 to the 12 is the number decaying each second, and it is offered as an option; the question asks for the number over 60 s. The other trap is multiplying without checking that the interval is short compared with the half-life.

Q13 · An estimate of a daily volume: the discriminator is that every conversion from a rate per minute to a total per day must be carried out — **Answer E**

What is being tested. Whether you carry out every conversion. The volume per breath and the rate are both given, and the only work is turning a rate per minute into a total per day.

Step 1 — what is the answer the product of?

$$\text{volume} = (\text{volume per breath}) \times (\text{breaths per day})$$

Step 2 — the rate, per hour.

$$12 \text{ breaths a minute} \times 60 \text{ minutes an hour} = 720 \text{ breaths an hour}$$

Step 3 — the rate, per day.

$$720 \times 24 = 17\,280 \text{ breaths a day}$$

Stopping here gives $720 \times 0.50 = 360$ litres. The boundary between 10 to the 2 and 10 to the 3 is $\sqrt{10} \times 10$ to the 2, which is about 320, and 360 is above it, so the hourly total alone would already round up to 10 to the 3. That is one of the options, and it is the answer to a question about an hour.

Step 4 — the volume.

$$17\,280 \times 0.50 = 8640 \text{ litres}$$

Step 5 — round. The boundary between 10 to the 3 and 10 to the 4 is $\sqrt{10} \times 10$ to the 3, which is about 3200, and 8640 is well above it, so the answer is 10 to the 4 litres.

A useful sanity check in litres: 8640 litres is 8.6 cubic metres, which is a cube about 2 m on a side. A person breathes out roughly their own volume of air every few minutes, which is the right order of magnitude and is the sort of check that catches a missed factor of sixty.

Answer: E

The trap. The trap is stopping one conversion short. Both 10 to the 2 and 10 to the 3 are reachable by doing part of the work, and they are the two most attractive wrong answers precisely because they come from a partial method rather than from a guess.

Q14 · Relative velocity of two bodies moving at right angles, obtained as a vector difference rather than a scalar one —

Answer C

What is being tested. Whether you treat velocity as a vector. The two speeds, 15 and 20, invite subtraction because that is what "relative" suggests, and subtraction gives 5.0, which is the first option.

Step 1 — the definition. The velocity of P relative to Q is what P's velocity looks like to an observer travelling with Q:

$$v_{PQ} = v_P - v_Q$$

Step 2 — draw the subtraction. Take the two velocity vectors nose to tail, as the figure does. The third side of the triangle, from the tail of the first to the head of the second, is the relative velocity. This is the step that turns an algebraic definition into a picture you can read.

Step 3 — the right angle. P travels east and Q travels north, and the roads meet at right angles, so the two velocity vectors are perpendicular. The triangle is right-angled, and the right-angle marker in the figure is what tells you so.

Step 4 — Pythagoras.

$$|v_{PQ}| = \sqrt{(15^2 + 20^2)} = \sqrt{(225 + 400)} = \sqrt{625} = 25 \text{ m s}^{-1}$$

Sanity check. The answer must lie between the difference of the two speeds and their sum, because that is what a vector difference can do: $20 - 15 = 5$ and $20 + 15 = 35$, and 25 is between them. A magnitude outside that range would be impossible. Since the two velocities are perpendicular, the answer is at the upper end of the range, which is where Pythagoras always puts it.

The 15-20-25 triangle is the 3-4-5 triangle scaled by five, which is why the answer is exact and why this question needs no calculator.

Answer: C

The trap. The trap is subtracting the magnitudes, which gives 5.0 and is the answer for two bodies moving along the same line in opposite directions. The perpendicularity is stated in the question and drawn in the figure, so both routes to the mistake have been closed off.

Q15 · A two-body system released from rest, where the driving mass reaches the floor and the sliding block is then decelerated by friction alone — **Answer D**

What is being tested. Whether you split the motion into two regimes and carry the speed across the boundary. The system is driven while B falls and undriven afterwards, and a single application of Newton's second law cannot cover both.

Step 1 — while the string is taut, the two blocks are one system. They are joined by an inextensible string, so they have the same acceleration. Treating them together removes the tension, which is internal and pulls in opposite directions on the two blocks.

Step 2 — the driving force. The only force driving the motion is the weight of the hanging block:

$$W_B = m_B g = 2.0 \times 10 = 20 \text{ N}$$

Step 3 — the retarding force. The block on the table feels friction. On a horizontal table the normal reaction equals its weight:

$$\begin{aligned} N &= m_A g = 3.0 \times 10 = 30 \text{ N} \\ f &= \mu N = 0.20 \times 30 = 6.0 \text{ N} \end{aligned}$$

Step 4 — the tension cancels. Writing the two equations separately, $20 - T = 2.0a$ for B and $T - 6.0 = 3.0a$ for A, and adding them gives $20 - 6.0 = 5.0a$. The tension has disappeared, which is why treating the two as a system is not a shortcut but the same algebra done in one line.

Step 5 — the acceleration.

$$a = (20 - 6.0)/5.0 = 14/5.0 = 2.8 \text{ m s}^{-2}$$

Step 6 — the speed when B lands. B falls 1.0 m, and A moves 1.0 m in the same time because the string does not stretch. Starting from rest:

$$v^2 = 2as = 2 \times 2.8 \times 1.0 = 5.6$$

Step 7 — the second regime. B has landed, so the string is slack and no longer pulls A. The only horizontal force on A is friction, acting backwards.

Step 8 — the deceleration. Newton's second law for A alone:

$$a' = f/m_A = 6.0/3.0 = 2.0 \text{ m s}^{-2}$$

Equivalently, $a' = \mu g = 0.20 \times 10 = 2.0 \text{ m s}^{-2}$. The mass cancels, which is why the stopping distance of a sliding object does not depend on how heavy it is.

Step 9 — the stopping distance. Using $v^2 = 2a's$ again, this time with the final speed zero:

$$s = v^2/(2a') = 5.6/(2 \times 2.0) = 5.6/4.0 = 1.4 \text{ m}$$

Sanity check. The deceleration, 2.0 m s^{-2} , is smaller than the acceleration, 2.8 m s^{-2} , so the second stage must be longer than the first: A accelerated over 1.0 m and coasts to a stop over 1.4 m. If the deceleration had come out larger than the acceleration, the stopping distance would have had to be shorter than 1.0 m, and an answer above it would have been wrong on that ground alone.

Note also that the stopping distance is the answer to the question as asked. The total distance A travels is 2.4 m, which is not asked for; the 1.0 m it travels while B is still falling is behind it by the time the string goes slack.

Answer: D

The trap. Two traps. The first is using the whole 2.4 m of travel in the second stage, which double-counts the driven part. The second is forgetting that friction continues after the string goes slack: a candidate who assumes A stops as soon as the drive is removed reports no further travel at all, or divides by the wrong acceleration.

Q16 · The greatest speed round a flat bend, obtained by equating the friction limit to the centripetal force and giving the answer as an expression — **Answer A**

What is being tested. Whether you know which force supplies the centripetal force on a flat bend, and whether you can cancel the mass before trying to solve for the speed. The answer is an expression, so the work is algebra rather than arithmetic.

Step 1 — what is pushing the car towards the centre? The car is on a flat road, so there is no component of the weight towards the centre and no banked surface to help. The only horizontal force is friction between the tyres and the road, and it must point radially inward. That is what the red arrow in the figure shows. A candidate who draws the friction along the direction of travel has no centripetal force at all.

Step 2 — the force needed. For motion in a circle of radius r at speed v :

$$F = m v^2 / r$$

Step 3 — the force available. The friction has a maximum, reached when the tyres are on the point of sliding:

$$f_{\text{max}} = \mu N = \mu mg$$

On a flat road the normal reaction is the weight. On a banked road it would not be, which is why this question specifies flat.

Step 4 — equate and cancel.

$$m v^2 / r = \mu mg$$

The mass appears on both sides and cancels. This is worth pausing on: it means the answer does not depend on whether the vehicle is a bicycle or a lorry. A heavier vehicle needs a larger centripetal force, and it also has a proportionally larger friction limit, so the two effects cancel exactly.

Step 5 — solve for the speed.

$$v^2 = \mu gr \quad \text{so} \quad v = \sqrt{(\mu gr)}$$

Sanity check. The speed rises with the square root of the radius, so a bend of four times the radius can be taken at twice the speed. It rises with the square root of the coefficient, so doubling the grip multiplies the speed by 1.4 rather than by two. Both of those are things one can check by driving, and both are consistent with everyday experience.

With the numbers of a typical bend, $\mu = 0.60$, $r = 50$ m and $g = 10$ N kg⁻¹, the speed is $\sqrt{300} = 17$ m s⁻¹, which is about 60 km per hour. That is the right order for a wet roundabout, which is a further check on the expression.

Answer: A

The trap. The trap is the mass. It looks as though the answer must depend on m , and an option that keeps m would be tempting; but the mass cancels, and the fact that it does is the physical content of the question. The second trap is forgetting the square root, which leaves μgr , the square of the speed.

Q17 · Linear thermal expansion of a rail, using the original length and a temperature difference in kelvin —

Answer E

What is being tested. Whether you take the temperature DIFFERENCE rather than the final temperature, and whether you keep track of the powers of ten. The arithmetic is a single multiplication.

Step 1 — the relation. The increase in length is

$$\Delta \ell = \ell_0 \alpha \Delta T$$

The length is the *original* length, not the expanded one. The difference matters at the level of a thousandth of a per cent here, but writing the relation correctly is what makes it not matter at all.

Step 2 — the temperature rise.

$$\Delta T = 120 - 20 = 100 \text{ K}$$

The coefficient is quoted per kelvin, and a temperature *difference* of 100 degrees Celsius is a difference of 100 kelvin. This is the one place where the two scales can be mixed without error, and it is worth knowing exactly why: the offsets cancel when you subtract. Using the final temperature, 120 °C, as the rise would be the mistake that gives 4.8 mm.

Step 3 — the expansion per kelvin.

$$\ell_0 \alpha = 2.0 \times 1.2 \times 10^{-5} = 2.4 \times 10^{-5} \text{ m per kelvin}$$

Step 4 — the expansion.

$$\Delta \ell = 2.4 \times 10^{-5} \times 100 = 2.4 \times 10^{-3} \text{ m} = 2.4 \text{ mm}$$

Sanity check. The rail has grown by 2.4 mm in 2.0 m, which is about one part in a thousand. This is the check that matters, because the whole relation $\Delta \ell = \ell_0 \alpha \Delta T$ assumes the coefficient is constant over the range and that the expansion is a small perturbation. Both are true here. If the answer had come out as centimetres, the linear relation itself would have been in doubt.

Two millimetres on a two metre rail sounds negligible until you consider that rails are joined end to end: over a kilometre of track the same expansion amounts to more than a metre, which is why real track is laid with expansion gaps.

Answer: E

The trap. The trap is using the final temperature rather than the rise. The question gives two temperatures, 20 and 120, and a candidate who reads the 120 as the rise gets exactly twice the answer. The other trap is the powers of ten: the coefficient is 10 to the minus 5 and the expansion is wanted in millimetres, which is 10 to the minus 3, so two conversions must be kept straight.

Q18 · The Doppler shift for a source receding from a stationary observer, expanded to first order in the ratio of the speeds — **Answer B**

What is being tested. Whether you know the sign of the Doppler correction, and whether you can expand a reciprocal to first order. The phrase "to first order in v/u " tells you the expansion is expected.

Step 1 — what must the answer do? The source is moving away. Each wavefront is emitted from a point slightly further from the observer than the last, so the waves arrive stretched out and the observed frequency is lower than the emitted one. Whatever expression you choose, it must give a frequency below f .

Step 2 — the exact relation. For a source receding at speed v in a medium where sound travels at u :

$$f' = f u / (u + v)$$

Dividing the top and bottom by u puts it in terms of the single small ratio:

$$f' = f / (1 + v/u)$$

Step 3 — expand. For a small quantity x , $1/(1 + x)$ is approximately $1 - x$. Here $x = v/u$, which the question states is small, so

$$f' \approx f(1 - v/u)$$

Sanity check. The factor is below one, so the observed frequency is below the emitted one, which is what a receding source must give. As a further check, if v were zero the factor would be exactly one and the observed frequency would equal the emitted one, which is right.

The expansion also shows why the phrase "to first order" matters. The exact factor is $1 - x + x^2 - \dots$, so the next term is a hundredth of a per cent for a ratio of one per cent. For sound from a car the ratio is of order 0.03, and the second-order term is a thousandth of the first, which is why the linear form is the one worth remembering.

Why the other options fail. $f(1 + v/u)$ and $fu/(u - v)$ both raise the frequency, so both describe an approaching source. $f(1 - u/v)$ inverts the ratio: when v is small, u/v is large, so the correction would be enormous, which contradicts the assumption the question makes. And $f(1 + v^2/u^2)$ is second order, so it misses the leading term entirely.

Answer: B

The trap. The trap is the sign. A candidate who remembers only that the Doppler effect changes the frequency, without remembering which way, is choosing between five options that differ mainly in sign. The physical check — a receding source is heard lower — settles it before any algebra is done.

Q19 · Apparent depth of an object seen through a transparent liquid, found by dividing the real depth by the refractive index — **Answer C**

What is being tested. The direction of the relation. Apparent depth is the real depth divided by the refractive index, so the bottom of a pool always looks closer than it is. Getting the division the wrong way round gives a tank that looks half as deep again, which is the first option.

Step 1 — why the image moves. A ray from the stone travels up through the liquid and bends away from the normal as it enters the air, because it is passing into a medium in which light travels faster. The eye does not know the ray has bent; it traces the ray backwards in a straight line. That backwards line meets the vertical through the stone at a point above the stone. That point is where the stone appears to be.

Step 2 — the relation. For small angles, so that the sines may be replaced by the tangents, the geometry gives

$$\text{apparent depth} = \text{real depth} / n$$

The index divides. This is worth remembering as a physical statement rather than a formula: a pool looks shallower than it is, so the apparent depth must be the smaller number, and the only way to get a smaller number from a depth and an index greater than one is to divide.

Step 3 — substitute.

$$\text{apparent depth} = 3.0 / 1.5 = 2.0 \text{ m}$$

Sanity check. The answer is two thirds of the real depth, which is a substantial shift: the stone appears a full metre higher than it is. That is consistent with everyday experience of looking into water, and it is why spearfishing is difficult. The answer is also smaller than the real depth, as it must be.

The figure shows the apparent image between the object and the surface, which is the geometric statement of the same thing. If your working put the image below the stone, the figure tells you immediately that something has been inverted.

Answer: C

The trap. The trap is multiplying instead of dividing, which gives 4.5 m and describes a pool that looks deeper than it is. The physical check — water always looks shallower than it is — rules that out before any formula is used.

Q20 · A series resistor in front of a parallel pair: finding the current in one branch of the pair — **Answer D**

What is being tested. Whether you find the p.d. across the parallel pair before dividing by a branch resistance. Using the full 6.0 V across the 2.0 Ω gives 3.0 A, which is far too large; using the cell current as the branch current gives 1.3 A.

Step 1 — combine the pair.

$$R_{\text{pair}} = 6.0 \times 2.0 / (6.0 + 2.0) = 12/8 = 1.5 \text{ } \Omega$$

The pair is smaller than either of its members, which is a check worth making every time.

Step 2 — the circuit resistance. The pair is in series with the 3.0 Ω :

$$R = 3.0 + 1.5 = 4.5 \, \Omega$$

Step 3 — the current from the cell. The internal resistance is negligible here, so the whole 6.0 V appears across the external circuit:

$$I = 6.0/4.5 = 1.3 \, \text{A}$$

Step 4 — the p.d. across the pair. This is the step that decides the question. The pair does not sit across the full 6.0 V, because the $3.0 \, \Omega$ takes $1.3 \times 3.0 = 4.0 \, \text{V}$ of it first:

$$V_{\text{pair}} = 1.3 \times 1.5 = 2.0 \, \text{V}$$

As a check, $4.0 \, \text{V} + 2.0 \, \text{V} = 6.0 \, \text{V}$, which is the whole supply. The p.d.s in a series circuit must add to the supply voltage, and here they do.

Step 5 — the branch current.

$$I(2.0 \, \Omega) = 2.0/2.0 = 1.0 \, \text{A}$$

Sanity check. The other branch carries $2.0/6.0 = 0.33 \, \text{A}$, and $1.0 + 0.33 = 1.3 \, \text{A}$, which is the current from the cell. Every junction must balance, and this one does.

Notice which branch carries more. The $2.0 \, \Omega$ carries three times the current of the $6.0 \, \Omega$, because the currents in a parallel pair divide in the inverse ratio of the resistances. A candidate who splits the 1.3 A in the ratio 6 : 2 instead of 2 : 6 gets 0.33 A and 1.0 A the wrong way round, which is exactly the last option.

Answer: D

The trap. Two traps. The first is using the supply voltage across the pair, which gives 3.0 A. The second is splitting the cell current in the ratio of the resistances rather than the inverse ratio, which gives the smaller share to the smaller resistance. The figure marks the junctions so that which pair is in parallel is unambiguous.

Q21 · Charge sharing between two capacitors: charge is conserved but energy is not, so a fraction of the energy is dissipated — **Answer A**

What is being tested. Whether you know which quantity is conserved. Charge is; energy is not. A candidate who assumes both are conserved reaches the fourth option and has to believe that no energy was lost, which the resistance of the connecting wires rules out.

Step 1 — what is conserved? The two capacitors are disconnected from the supply and joined to each other, so the charge has nowhere to go except from one plate to the other. The total charge is therefore unchanged. The energy is not: charge moves through wires that have resistance, and the current that flows heats them.

Step 2 — the charge before.

$$Q = C_1 V_1 = 4.0 \times 10^{-6} \times 10 = 4.0 \times 10^{-5} \, \text{C}$$

Step 3 — the common potential. After connection the two capacitors are in parallel and therefore share the same potential. Their capacitances add:

$$C_{\text{total}} = 4.0 + 6.0 = 10 \mu\text{F}$$

$$V = Q/C_{\text{total}} = 4.0 \times 10^{-5} / 10 \times 10^{-6} = 4.0 \text{ V}$$

The common potential is below the original 10 V, which is what you expect when the charge is spread over a larger capacitance.

Step 4 — the energy before and after.

$$E_{\text{before}} = \frac{1}{2} C_1 V_1^2 = \frac{1}{2} \times 4.0 \times 10^{-6} \times 100 = 2.0 \times 10^{-4} \text{ J}$$

$$E_{\text{after}} = \frac{1}{2} C_{\text{total}} V^2 = \frac{1}{2} \times 10 \times 10^{-6} \times 16 = 8.0 \times 10^{-5} \text{ J}$$

Step 5 — the fraction lost.

$$E_{\text{lost}} = 2.0 \times 10^{-4} - 8.0 \times 10^{-5} = 1.2 \times 10^{-4} \text{ J}$$

$$\text{fraction} = 1.2/2.0 = 0.60$$

So 60 % of the stored energy is dissipated. Note that the total charge afterwards is $10 \mu\text{F} \times 4.0 \text{ V} = 4.0 \times 10^{-5} \text{ C}$, which is exactly the charge we started with, so the conservation of charge is confirmed.

Sanity check. The answer must lie between zero and one, and it does. It also cannot be zero, because the final potential is 4.0 V rather than 10 V and the energy stored falls with the square of the potential: had the energy been conserved, the final potential would have had to be higher than the initial one, which the charge calculation rules out.

It is worth noticing that the fraction lost does not depend on the initial voltage. Connecting a $4.0 \mu\text{F}$ to a $6.0 \mu\text{F}$ always loses 60 % of the energy, whatever the starting potential, because both the before and the after energy scale with the square of it.

Answer: A

The trap. The trap is assuming energy is conserved. Charge sharing is not an elastic process: the current that redistributes the charge dissipates energy in the resistance of the wires, and there is no way to connect two capacitors without it. The 0 % option is the misconception stated plainly.

Q22 · Specific latent heat of fusion from the length of the plateau on a heating curve at constant power —

Answer C

What is being tested. Whether you know what a plateau means, and whether you use the right time from the graph. The plateau is the melting stage; the 4.0 min before it is the warming stage, and using that time instead is the first option.

Step 1 — read the graph. The temperature rises from 20 °C to 80 °C over the first 4.0 min. It then stays at 80 °C from 4.0 min to 10 min, which is a plateau 6.0 min long. Then it rises again. A constant temperature while energy is being supplied means the energy is being used to change the state, not to raise the temperature. That is the definition of a latent heat.

Step 2 — the plateau's length in seconds.

$$t = 6.0 \text{ min} = 6.0 \times 60 = 360 \text{ s}$$

Step 3 — the energy supplied during melting. The power is constant, so

$$E = P t = 150 \times 360 = 54\,000 \text{ J}$$

Step 4 — the specific latent heat. The specific latent heat of fusion is the energy needed to melt one kilogram at the melting point:

$$L = E/m = 54\,000/0.30 = 180\,000 \text{ J kg}^{-1}$$

That is $1.8 \times 10^5 \text{ J kg}^{-1}$.

Step 5 — the check the graph gives you for free. The warming stage supplies $150 \times 240 = 36\,000 \text{ J}$, and the plateau supplies $54\,000 \text{ J}$. The plateau is half again as long as the warming stage, and it supplies half again as much energy. The two facts agree, which they must because the power is constant. If your answer had used the 4.0 min you would have got 1.2×10^5 , which is the warming stage's energy per kilogram rather than the latent heat.

Sanity check. The specific latent heat of fusion of ice is $3.3 \times 10^5 \text{ J kg}^{-1}$, and of lead about 2.3×10^4 . The value here, 1.8×10^5 , sits between them, which is the right region for a solid with moderate intermolecular forces. A value of 10^7 would be a hundred times ice and would signal an arithmetic slip.

Answer: C

The trap. The trap is using the warming stage's time for the melting stage. Both stages are on the same graph and both are labelled, so the only defence is to know that the plateau is the one that carries the latent heat. A second trap is forgetting to convert minutes to seconds, which gives an answer sixty times too small.

Q23 · The fraction of a radioactive sample remaining after a whole number of half-lives — **Answer B**

What is being tested. Whether you halve repeatedly rather than dividing. After one half-life half remain; after two, a quarter; after three, an eighth.

Step 1 — how many half-lives?

$$24 \text{ days} / 8.0 \text{ days per half-life} = 3 \text{ half-lives}$$

Step 2 — halve three times.

$$1 \rightarrow 1/2 \rightarrow 1/4 \rightarrow 1/8$$

The fraction remaining is $1/8$.

Sanity check. The answer must be a power of one half, because each half-life multiplies the surviving fraction by one half. It also must not be zero: after three half-lives there is still an eighth of the sample, and after ten there is still a thousandth. Decay never quite finishes, which is why the question asks for a fraction rather than for "how many are left".

Answer: B

The trap. The trap is dividing 1 by 3 rather than raising one half to the third power. The two operations differ because decay is exponential: the loss in each half-life is half of what is left at the time, not half of the original sample.

Q24 · Threshold wavelength for the photoelectric effect, obtained by equating the photon energy to the work function

— **Answer E**

What is being tested. Whether you can set up the threshold condition and rearrange it, and whether you can check the result by dimensions when the answer is an expression rather than a number.

Step 1 — the photon energy in terms of wavelength. A photon of frequency f carries energy hf , and since $c = f\lambda$ the frequency is c/λ . So

$$E = hf = hc/\lambda$$

Longer wavelength means less energy. That is the fact the word "longest" is pointing at.

Step 2 — the threshold condition. The work function is the least energy that will remove an electron. At the threshold the photon has exactly that much and the electron is released with no kinetic energy at all:

$$hc/\lambda = \phi$$

If the wavelength were longer, the photon would carry less than ϕ and no electron would come off. So this equation defines the longest wavelength that works.

Step 3 — rearrange.

$$\lambda = hc/\phi$$

Step 4 — check the dimensions. The planck constant has the dimensions of energy times time, the speed of light is a length over a time, and the work function is an energy. So hc/ϕ has dimensions of (energy $\times$ time $\times$ length / time) / energy, which is a length. The expression is a wavelength. That single check rules out $\phi/(hc)$, which is a reciprocal length, and $hc\phi$, which is an energy squared times a time.

Sanity check. A metal with a large work function should need short wavelengths, since each photon must carry more energy. The expression hc/ϕ falls as ϕ rises, which is the right direction.

A numerical check. Using $hc \approx 2.0 \times 10^{-25} \text{ J m}$, a work function of $3.0 \times 10^{-19} \text{ J}$ gives a threshold wavelength of $2.0 \times 10^{-25}/3.0 \times 10^{-19}$, which is about $6.6 \times 10^{-7} \text{ m}$. That is 660 nm, which is red light. A metal with a work function of $3.0 \times 10^{-19} \text{ J}$ therefore responds to red and everything bluer, and not to infrared. That is the right order for a real photocathode, which is a check on the whole setup.

Answer: E

The trap. The trap is the arrangement of the three symbols. Every option is built from the same h , c and work function, so the only way to be sure is to derive the relation and then check its dimensions. Multiplying all three, or dividing by hc , gives something that is not a length.

Q25 · A floating block: the submerged fraction is the ratio of the block's density to the liquid's, so the mass follows from the displaced volume — **Answer D**

What is being tested. Whether you use the submerged fraction rather than its complement, and whether you end with a mass rather than a weight. The answer is an expression, so the work is setting up the equilibrium and cancelling g .

Step 1 — the equilibrium condition. The block floats without accelerating, so the upward force on it equals its weight:

$$\text{upthrust} = mg$$

Step 2 — the upthrust. The upthrust on a floating body is the weight of the liquid it has pushed aside. The block is approximately uniform, so the displaced fraction is the same at every depth, and the displaced volume is the part below the surface:

$$\begin{aligned}\text{displaced volume} &= 0.60V \\ \text{upthrust} &= \rho_w \times 0.60V \times g\end{aligned}$$

Step 3 — equate and cancel.

$$mg = 0.60 \rho_w V g$$

The g appears on both sides and cancels, which is why the answer does not contain it. This is worth noticing: a floating body's behaviour does not depend on the strength of gravity, so the same block floats to the same depth on the Moon.

Step 4 — the mass.

$$m = 0.60 \rho_w V$$

Sanity check. The mass of the block is 0.60 of the mass of the same volume of water. That is less than $\rho_w V$, which it must be, because a body that is denser than the liquid sinks. It is also more than $0.40 \rho_w V$, because the block is more than half submerged and therefore more than half as dense as water.

The result generalises in a way worth remembering: the submerged fraction equals the ratio of the body's average density to the liquid's. Here 0.60 submerged means the wood has an average density of $0.60 \rho_w$, which for water is 600 kg m^{-3} . That is the density of dry pine, which is a good sign that the numbers in the question are the ones a real block would have.

Answer: D

The trap. Two traps. The first is using the 0.40 above the surface, which gives the complement of the right answer and is the mistake the figure is drawn to catch by labelling both parts. The second is ending with the weight, $\rho_w V g$, rather than the mass.